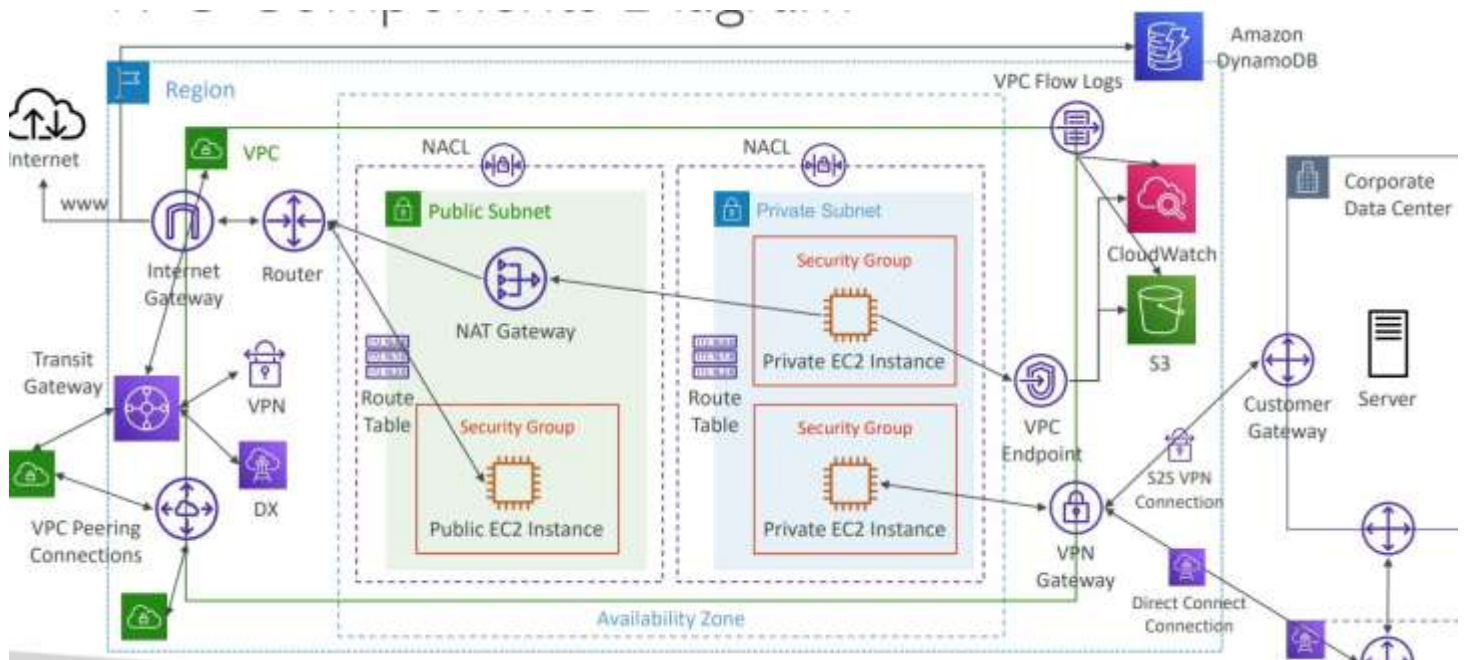


Creating an AWS VPC from Scratch (ENTERPRISE GRADE)



This is the final VPC which I will construct step by step

VPC

Stands for Virtual Private Cloud

In this guide, I'll create a custom VPC in AWS from scratch, including subnets, route tables, Internet connectivity, and essential components. This provides better control and security than using the default VPC.

This VPC **WILL BE** an Enterprise grade VPC with options and features which ensure

1. High Availability and scalability
2. Resilience
3. Security
4. High performance and traffic spiking workloads

Let's make it from scratch

Step1.

Creating a VPC and establishing subnets in a **region**

VPC as stated is a virtual private cloud in short like an independent society with residents not dependent on the outside for any resources.

Imagine Subnets like the houses in the Neighborhood Some our public like movie halls and parks

Rest are private like residential Homes



The screenshot shows the AWS Management Console 'Subnets' page. The page displays a list of subnets with columns for Subnet ID, State, VPC, Block Public IP, IPv4 CIDR, and IPv6 CIDR. The subnets are all in an 'Available' state.

Subnet ID	State	VPC	Block Public IP	IPv4 CIDR	IPv6 CIDR
subnet-087e486d2b3f9ce5d	Available	vpc-05d1f5e9b9e9c526e dem...	Off	10.0.1.0/24	-
subnet-0f9e0fa1c50402b5d	Available	vpc-02c0dch56c281e19c9 defa...	Off	172.31.0.0/20	-
subnet-096d3b467ce2bcb8f	Available	vpc-02c0c056c281e19c9 defa...	Off	172.31.16.0/20	-
subnet-09d288c305339647	Available	vpc-05ef1f5a9b9e9c526e dem...	Off	10.0.0.0/24	-
subnet-07a84d294cc1aaa07	Available	vpc-02c0dch56c281e19c9 defa...	Off	172.31.32.0/20	-
subnet-0c548e7b3005cd8f6	Available	vpc-05ef1f5a9b9e9c526e dem...	Off	10.0.2.0/24	-

VPC [Show details](#)

Your AWS virtual network

demo-vpc

Subnets (3)

Subnets within this VPC

ap-south-1a
newBabySubnet

ap-south-1b
babysubnet2

ap-south-1c
newbabysubnet3

STEP2

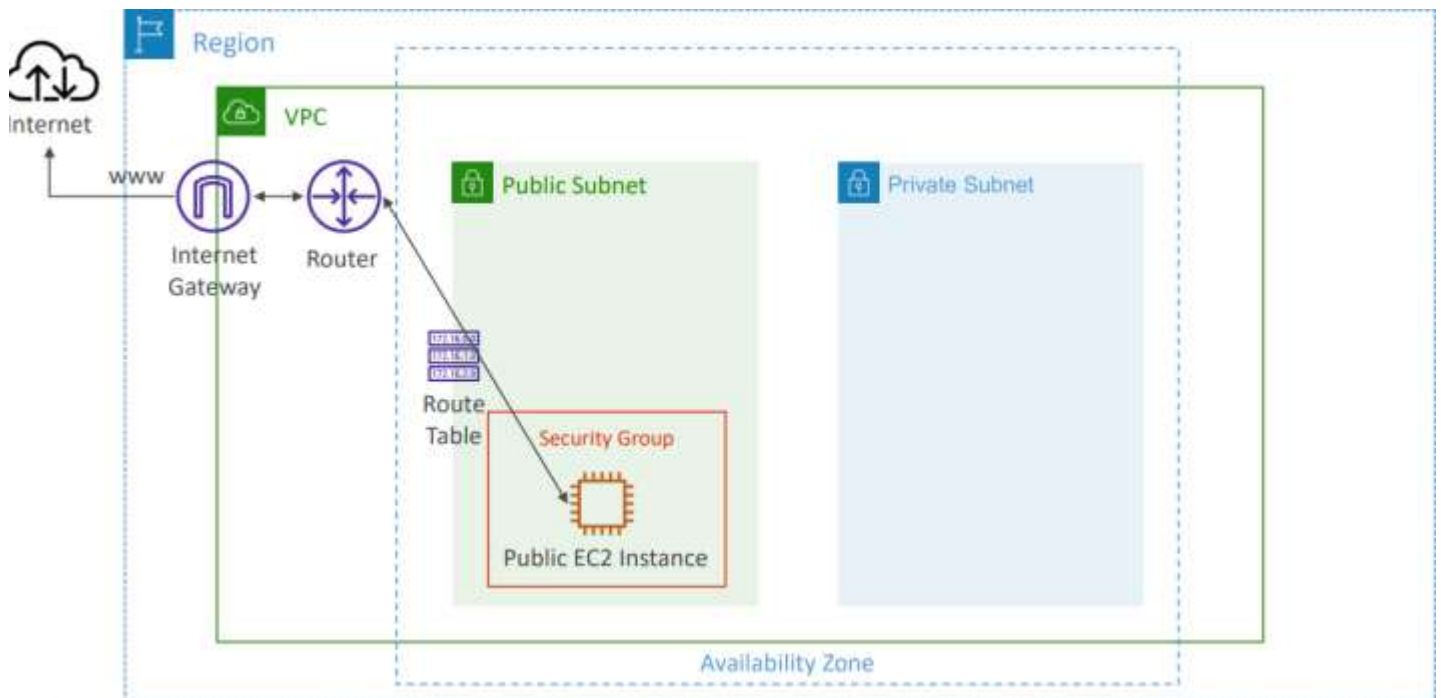
Internet GateWay

Now that the neighborhoods created, we require a connection for import exports or in cloud terms Internet Access to the instances containing the app

I created an internet Gateway now the diagram is like this

Note:

- It scales horizontally and is highly available and redundant
- Must be created separately from a VPC
- One VPC can only be attached to one IGW and vice versa
- Internet Gateways on their own do not allow Internet access...
- Route tables must also be edited!



Below shows the attached Internet Gateway
and the route table records being attached to the Internet Gateway

PUBLIC SUBNET	PRIVATE SUBNET
Its personal Route table is created pointing the Internet Gateway	Its personal Route table is created either pointing to a Bastion Host or private networks only

igw-0ff212bc85a35b62e / demoInternetGateway

Details

Internet gateway ID: igw-0ff212bc85a35b62e State: Attached VPC ID: vpc-03d115a9b8e9c526e (demo-vpc) Owner: 254159011250

Tags

Key	Value
Name	demoInternetGateway

Routes (3)

Destination	Target	Status	Propagated
0.0.0.0/0	igw-0ff212bc85a35b62e	Active	No
10.0.0.0/16	local	Active	No
172.31.0.0/16	pcx-00ca29223482e24c7	Active	No

So now you might be having a doubt?

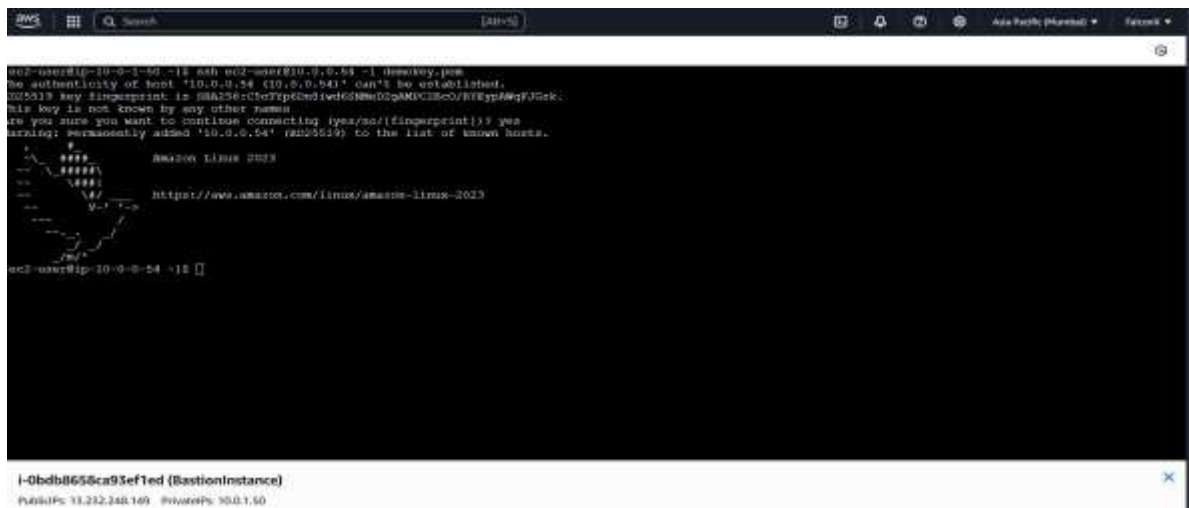
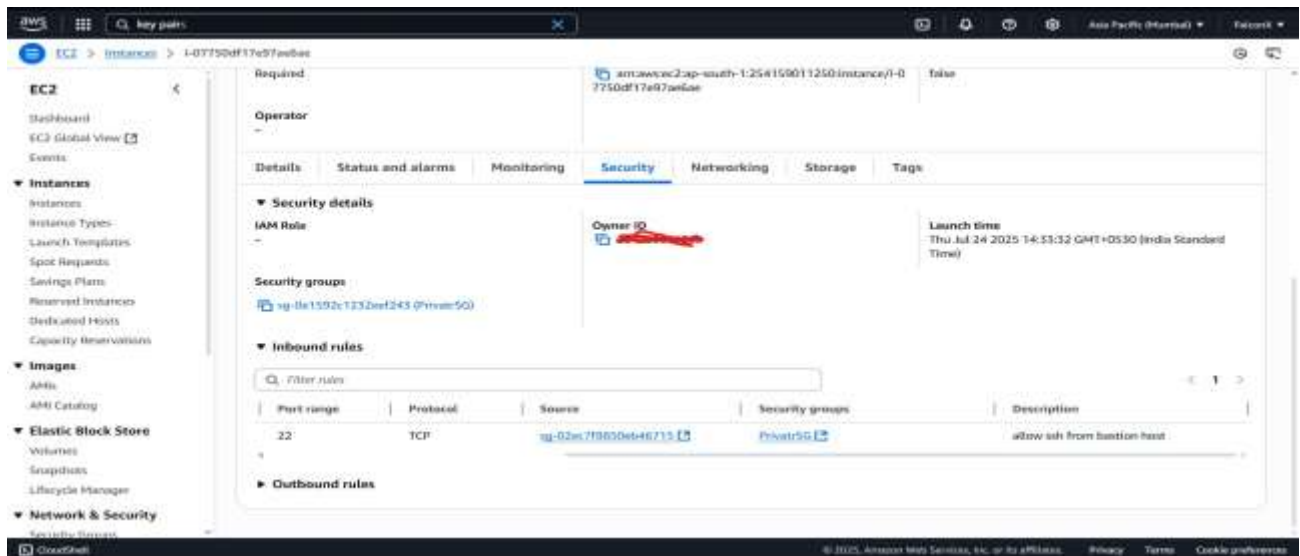
I want to keep my applications private so that they are visible only to me. For those, I will not give a connection to the Internet Gateway. So, from where will they get the network?"

Bastion Hosts (Step-2.5)

These are the types of instances which are hosted on public subnets have access and are visible to the public and can operate resources to the private subnet instances simply but doing SSH into those instances

Here I have

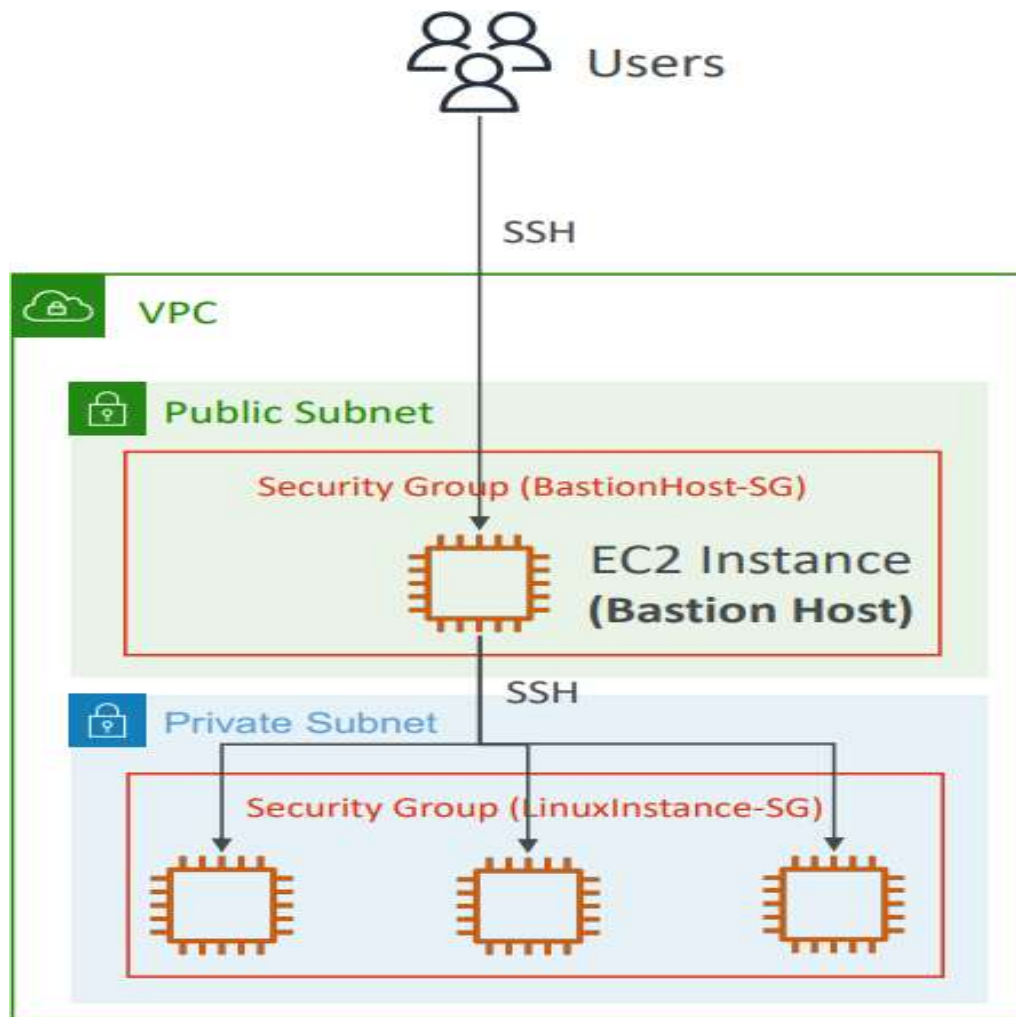
1. Created instance amid bastion host in public subnet and one in private subnet
2. Modified the security group inbound rules of private instance to allow traffic on port 22(SSH Port) to point FROM THE SECURITY GROUP OF BASTION HOST
3. Now I ssh into the Bastion host
4. From here I ssh into the private instance – make a .pem file ,import it in the terminal itself and echo a “Hello” in Private instance
5. As soon as I curl cmd(For calling) it in the public instance it returns the desired output



Expected Output:

```
[ec2-user@ip-172-31-3-7 ~]$ curl 10.0.1.50  
hello  
[ec2-user@ip-172-31-3-7 ~]$
```

NOW THE Structure LOOK LIKE THIS



Step 3

Increasing level

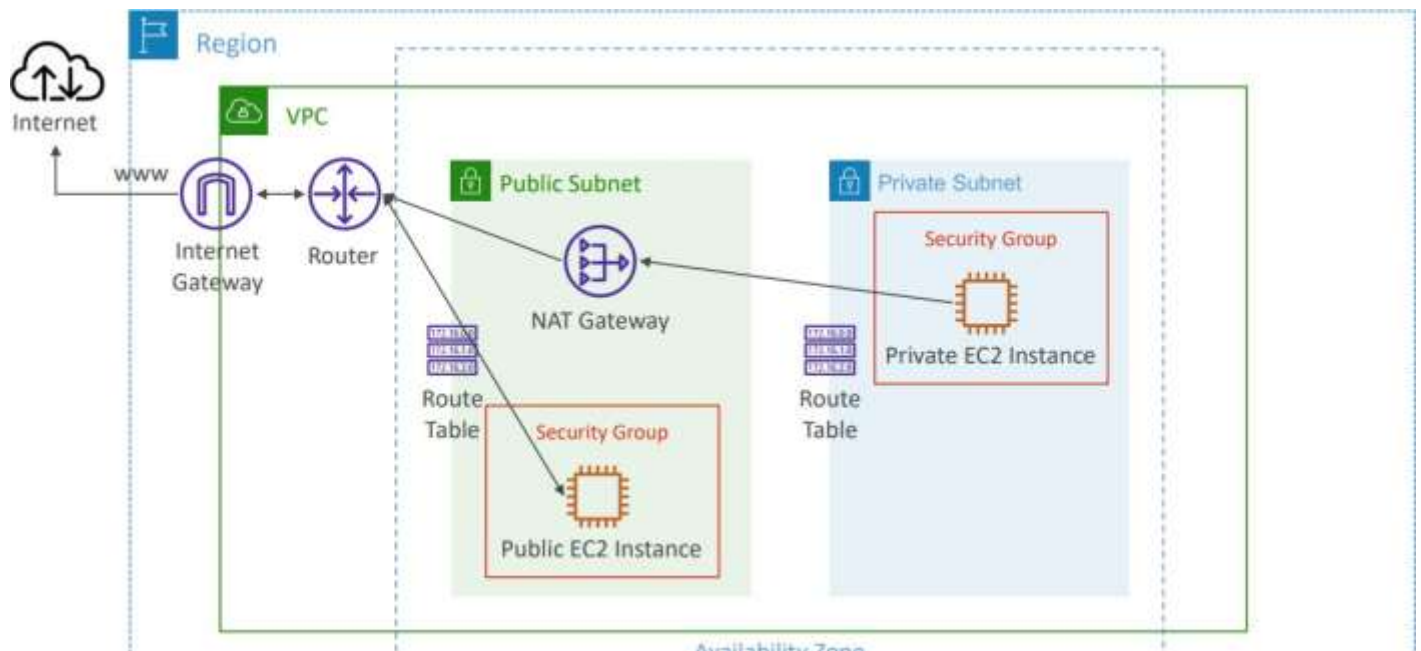
This method is great but not reliable for bigger workloads so

NAT GATEWAY

This thing ensures a stable internet connection both in and out be given to private subnets with a window for security checks of!

AWS-managed NAT, higher bandwidth, high availability, no administration

- NATGW is created in a specific Availability Zone, uses an Elastic IP
- Can't be used by EC2 instance in the same subnet (only from other subnets)
- Requires an IGW (Private Subnet => NATGW => IGW)
- 5 Gbps of bandwidth with automatic scaling up to 100 Gbps



Now Comes the Security Part to secure the infrastructure for

1. Malicious Traffic
2. Misguided Traffic

Step-4

NACL(s) /Security Groups

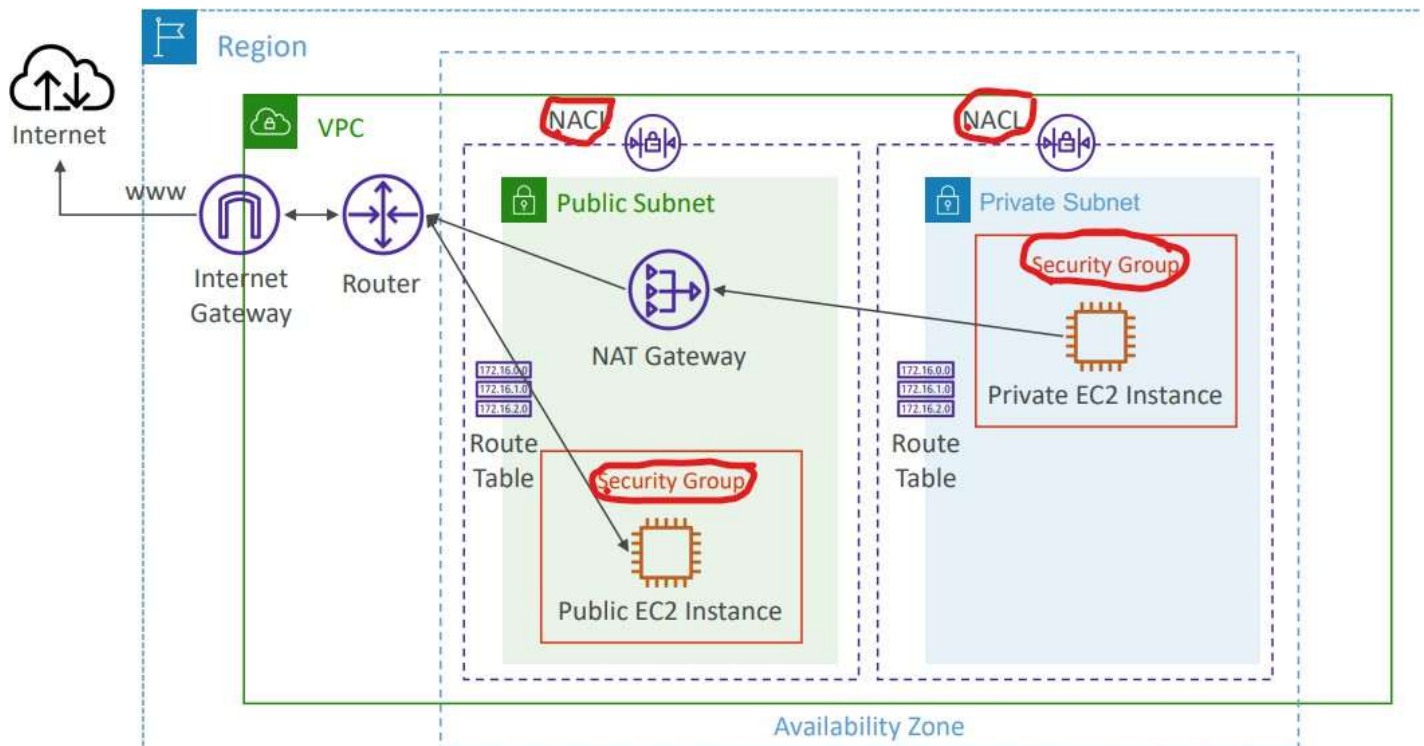
NACL(s) are Subnet Level completely stateless (Check both in and out traffic) and are based on Rules priority no's.

Security groups are instance level to further prevent the misguided traffic to reach its correct path and may not activate a wrong instance these are stateful (logical people if in then out and vice versa types) and have no priority no. System just declarations

There are many places where I used NACLs to block certain CIDRs

And off those Which enter inside to guide through security groups

So here is our progress



NOTE:

EPHIMERAL PORTS- these are on spot usable ports used instead of defined ports for communication for added security.

Now If the app is Enterprise grade and the infrastructure is also Enterprise grade
It cannot rely on only 1 VPC

We have to have multiple VPC but that is not the point that is simple the point is

WE NEED SYNC and updated Information between all the VPCs for the companies

Also an Advantage of this will be to share info across VPC ,,,,next Feature, so that's
why we are using VPC peering.

Step 4

VPC Peering

It simply means establishing a peering connection or simply a connection between VPCs
CROSS ACCOUNT & CROSS REGION

Here I set up

1. TWO VPCs in different region
2. Accept the request of peering
3. **MOST IMP. ALWAYS** update it in the route tables which I will see in common once both VPCs are connected as shown

Accept VPC peering connection request

Info

Are you sure you want to accept this VPC peering connection request? (pcx-00ca29223482e24c7 / my-pc-1)

Requester VPC

vpc-03d1f5a9b8e9c526e / demo-vpc

Accepter CIDRs

—

Requester owner ID

254159011250(This account)

Accepter VPC

vpc-02c0cb56c281c19c9 / defaultVPC

Requester Region

Mumbai (ap-south-1)

Accepter owner ID

254159011250(This account)

Requester CIDRs

10.0.0.0/16

Accepter Region

Mumbai (ap-south-1)

Cancel

Accept request

VPC > Peering connections > Create peering connection

Select a local VPC to peer with

VPC ID (Requester)

vpc-02c0cb56c281c19c9 (defaultVPC)

VPC CIDRs for vpc-02c0cb56c281c19c9 (defaultVPC)

CIDR	Status	Status reason
172.31.0.0/16	Associated	-

Select another VPC to peer with

Account

☒ My account

☐ Another account

Region

☒ This Region (ap-south-1)

☐ Another Region

VPC ID (Accepter)

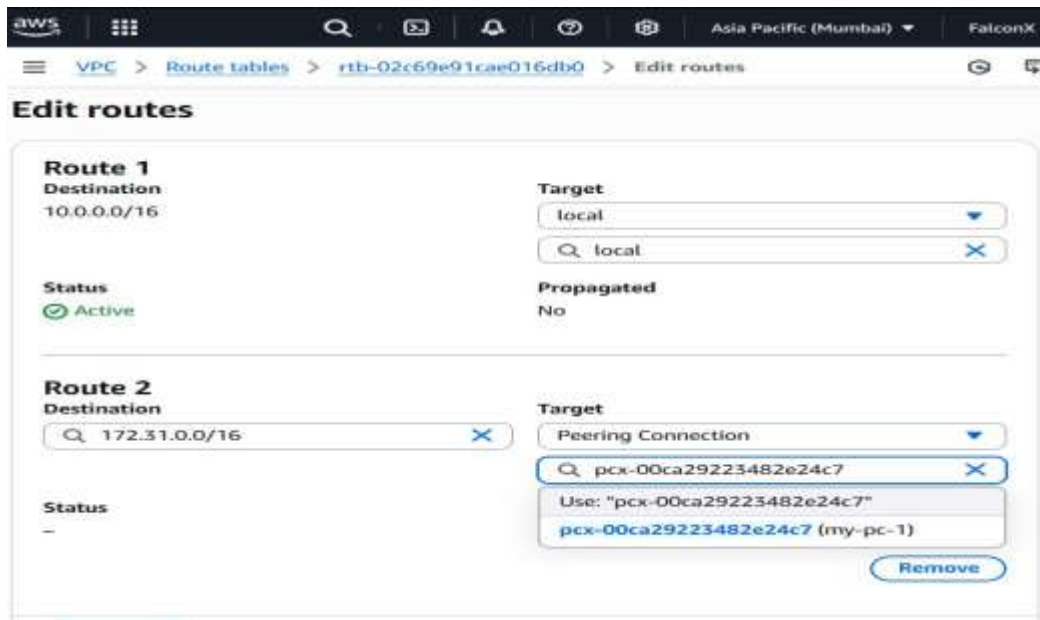
vpc-02c0cb56c281c19c9 (defaultVPC)

VPC CIDRs for vpc-02c0cb56c281c19c9 (defaultVPC)

CIDR	Status	Status reason
172.31.0.0/16	Associated	-

Tags

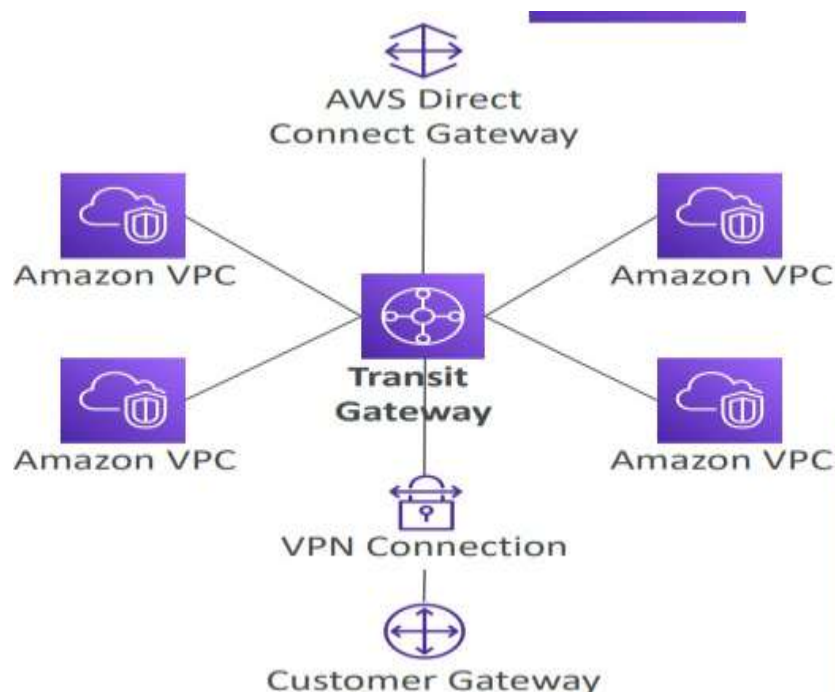
Changing the route Table



ADDITIONAL STEP FOR ENTERPRISE GRADE

Setup a Transit Gateway for incase of connecting multiple VPCs together using peering

Reason: Network Topology becomes **complicated** and difficult to maintain



Some Cost Optimizing approaches....

Okay So Now the thing is I want my Private Instance to access AWS services or my private DATA Center

Why should i incur such costs of setting up a NAT gateway then accessing those services through public

That's expensive and not even secure

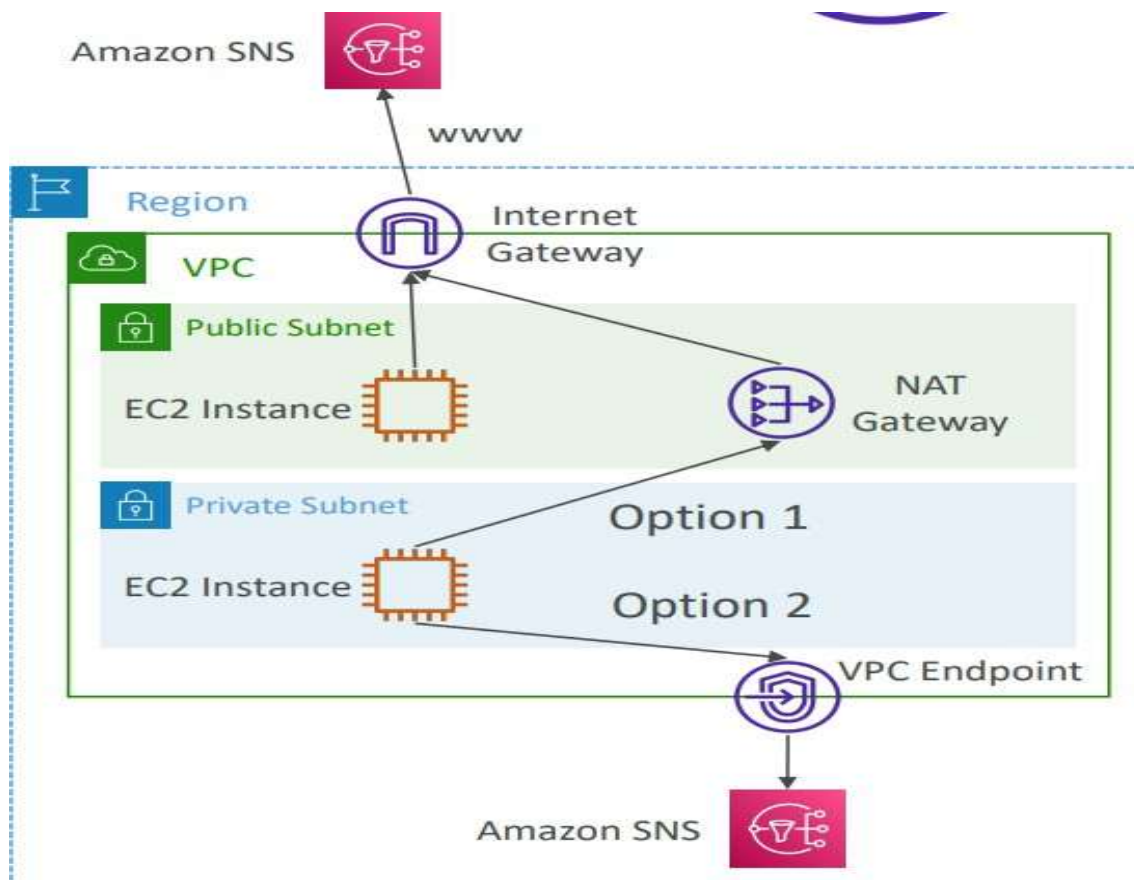
Hence

Step 5

VPC Endpoints

These are like metro connecting dots 🌐

VPC Endpoints (powered by AWS Private Link) allows you to connect to AWS services using a private network instead of using the public Internet



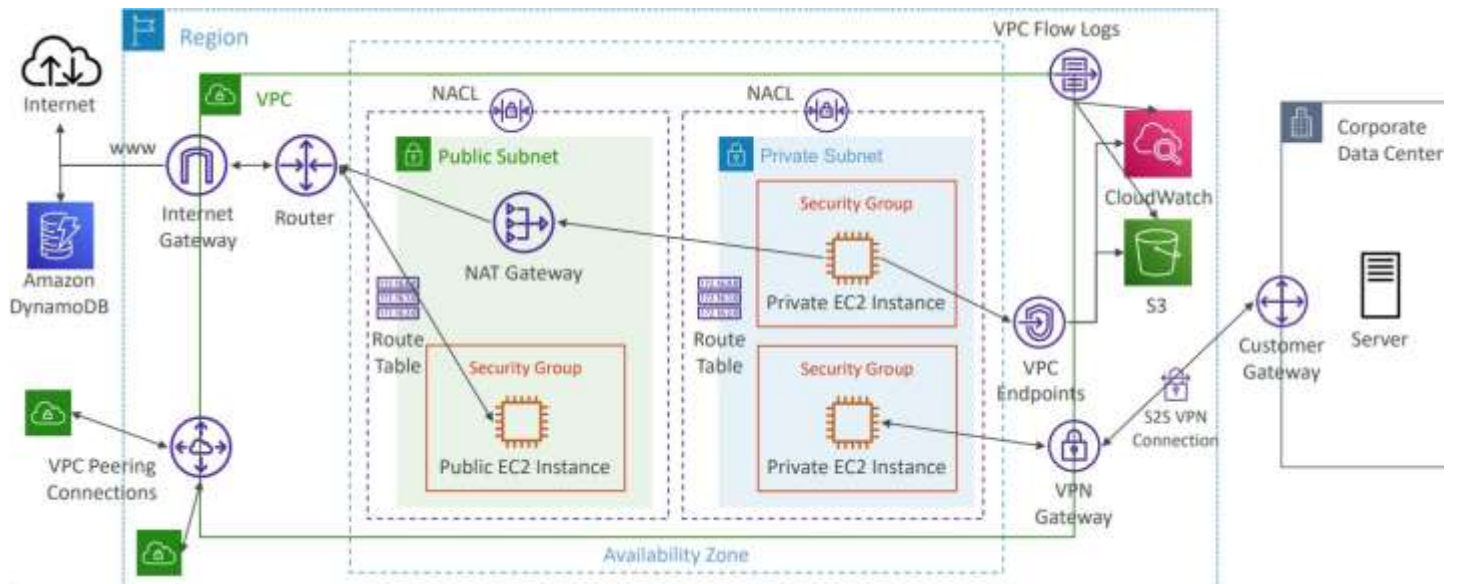
Now this is what I meant

These are also of two types one is a gateway endpoint supporting less services (FOR Small companies)

2. Is Interface endpoints (Basically Endpoint premium subscription for Enterprises)

Finally.... my VPC is all set !!!!

A final step remains.....



THANK YOU